Efficient Packet Routing for Large-Scale LEO Satellite Networks: A Pareto-Optimal MARL Approach With Queueing Theory

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***Abstract*—Low Earth orbit (LEO) satellite networks enhance terrestrial connectivity by providing global coverage and low-latency communication. However, their highly dynamic topology, time-varying propagation delays, and constrained bandwidth severely limit the efficiency of conventional centralized routing, underscoring the necessity for adaptive and distributed strategies that can operate effectively under partial observability. Multiagent reinforcement learning (MARL) offers a promising foundation for such strategies by enabling decentralized, context-aware decision-making based on local information. Nevertheless, existing MARL-based routing approaches often struggle to maintain accurate congestion awareness, reconcile conflicting objectives, and ensure stable convergence in large-scale LEO constellations. To address these challenges, we present pareto-optimal multi-agent proximal policy optimization packet routing framework (POMAP), a packet routing framework that integrates Pareto optimization with multiagent proximal policy optimization (MAPPO) to achieve efficient and stable tradeoffs across multiple key performance metrics. Specifically, we propose a dynamic multiattribute graph model for LEO satellite networks that simultaneously captures communication delay and energy consumption. Within this framework, each satellite node is represented as a G/G/1/K queue equipped with active queue management (AQM) and scheduled using weighted priority queueing (WPQ), thereby enabling precise characterization and control of packet queueing behavior. We formulate the packet routing problem as a partially observable Markov decision process (POMDP) that jointly minimizes delay, energy consumption, and packet loss rate, and apply MAPPO to optimize the resulting policy. Extensive simulations on realistic satellite network topologies demonstrate that the proposed method achieves better convergence stability, improves Pareto front coverage, and enhances overall network performance compared with the state-of-the-art baselines.**

***Index Terms*—Low Earth orbit (LEO) satellite networks, multiagent reinforcement learning (MARL), multiobjective optimization, packet routing.**

NOMENCLATURE

Notation	Definition/Explanation
$\mathcal{T}$	Set of time slots.
$G(t)$	Satellite network topology at time t.
V	Set of satellite nodes.
$E(t)$	Edges representing ISLs at time t.
$d_{ij}(t)$	Distance between satellites i and j at time t.
$L_{ij}(t)$	Free-space path loss (FSPL).
$h_{ij}(t)$	Channel gain between satellites i and j.
$\mathrm{SNR}_{ij}(t)$	Signal-to-noise ratio (SNR) between satellites i and j.
P_{tx}	Transmission power.
P_{rx}	Reception power.
N_0	Noise spectral density.
N_{total}	Total noise power = $N_0 B$.
$R_{ij}(t)$	Maximum data rate between satellites i and j.
S	Size of each packet (in bits).
$s_{ij}(t)$	Number of packets transmitted from node i to node j.
$u_{ji}(t)$	Number of packets received by node i from j.
$r_{i,m}(t)$	Number of arriving class-m packets at node i.
$x_{i,m}(t)$	Number of class-m packets served from queue m at node i during time slot t.
$q_{i,m}(t)$	Queue length of traffic class m at node i at time t.
$\mathcal{M}$	Set of traffic classes $\{1, 2, \ldots, M\}$.
$\mu_i(t)$	Total service rate at satellite i at time t.
$\mu_{i,m}(t)$	Service rate assigned to queue m.
K_m	Queue capacity for traffic class m.
K	Total queue capacity at a satellite.
$B_i(t)$	Set of backlogged queues at node i in the highest nonempty priority level.
$N_i(t)$	Set of neighboring satellites of node i at time t.
$w_{i,m}(t)$	Weighted priority queueing (WPQ) weight of queue m at node i.

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Δt	Time slot duration.
$D_{ij}^{\text{trans}}(t)$	Transmission delay between satellites i and j.
$D_{ij}^{\text{prop}}(t)$	Propagation delay between satellites i and j.
$D_{i,m}^{\text{queue}}(t)$	Queueing delay of class-m traffic at node i.
$D_{i}^{\text{queue}}(t)$	Aggregate queueing delay at node i.
$D_{ij}(t)$	Total communication delay between satellites i and j.
$a_{ij}(t)$	ISL availability from i to j.
$\alpha(t)$ and $\beta(t)$	Node-level normalized AQM thresholds (fractions of K_m).
$\delta_{1,m}(t)$	Lower queue threshold for class m under AQM.
$\delta_{2,m}(t)$	Upper queue threshold for class m under AQM.
$p_m(q_{i,m}(t))$	Drop probability for queue m based on queue length.
$p_{\max}(t)$	Maximum drop probability parameter (uniform across all queues at node i).
$p_{\max,m}(t)$	Maximum drop probability for class-m at node i (equal to $p_{\max}(t)$).
$E_i^{\text{tx}}(t)$	Transmission energy consumption at satellite i.
$E_i^{\text{rx}}(t)$	Reception energy consumption at satellite i.
$E_i(t)$	Total energy consumption at satellite i at time t.
E_i^{init}	Initial energy of satellite i.
$E_i^{\text{rem}}(t)$	Remaining energy at satellite i at time t.

I. Introduction

With the rapid advancement of communication technologies and the growing demand for global connectivity, satellite networks have become indispensable components of the next-generation mobile communication systems. Among them, low Earth orbit (LEO) satellites are particularly attractive, as they provide high-speed, low-latency global coverage and enable emerging applications such as the Internet of Things, autonomous vehicles, and smart cities [2]. Driven by major industrial initiatives from SpaceX, OneWeb, and Telesat, LEO constellations are expected to play a pivotal role in the evolution of sixth-generation (6G) communication systems [3], [4]. Nevertheless, LEO networks inherently suffer from highly dynamic topologies, variable propagation delays, and stringent bandwidth constraints [5], [6], which pose severe challenges to conventional terrestrial routing protocols that rely on stable connectivity and frequent, low-latency link-state updates.

A wide range of routing protocols and algorithms have been proposed in both academia and industry to cope with the unique characteristics of satellite networks [7]. Among them, centralized routing approaches are conceptually simple but suffer from poor scalability as constellation sizes expand. Specifically, such schemes impose heavy computational and storage burdens, require substantial bandwidth for frequent routing updates, and are vulnerable to single-point failures. In addition, computing globally optimal routes in large-scale LEO constellations is often computationally prohibitive since problems such as multicommodity flow optimization and dynamic shortest-path computation are generally intractable under realistic conditions. These limitations highlight the necessity of decentralized routing strategies that can operate efficiently under dynamic topologies and stringent resource constraints [8], [9].

Building on this motivation, fine-grained packet-level forwarding has emerged as a promising direction, as it can effectively adapt to dynamic and heterogeneous traffic patterns. Nevertheless, fully decentralized packet routing remains challenging since each node must select paths that jointly balance distance and congestion based only on local observations. To address this issue, many studies formulate packet routing as a decision-making problem under partial observability, which is often modeled as a partially observable Markov decision process (POMDP). Multiagent reinforcement learning (MARL) has shown considerable promise in solving such problems [10], [11], [12]. Building on this foundation, several MARL-based approaches have been specifically designed for satellite network routing [13], [14], [15].

Despite these advances, their performance in LEO satellite networks remains limited due to several unresolved issues. First, most approaches are designed for static terrestrial topologies and rely mainly on link-state metrics (e.g., bandwidth and latency), while overlooking critical congestion indicators such as queue occupancy and service capacity. This omission often delays congestion detection and reduces forwarding efficiency [16]. Second, agents are typically trained independently without parameter sharing, which weakens coordination and degrades performance at deployment time [17]. Third, reward design remains a major challenge: global rewards may lead to passive behavior, whereas overly local rewards risk encouraging selfish actions, both of which can harm overall system performance by slowing convergence and reducing resource utilization efficiency. Beyond these single-objective issues, practical satellite routing must also optimize multiple conflicting objectives, such as minimizing delay while conserving energy. Conventional reinforcement learning (RL) methods usually aggregate such objectives into a weighted sum, which risks introducing bias toward certain metrics. Multiobjective RL (MORL) alleviates this limitation by defining separate rewards for each objective [18], [19], while Pareto optimization provides balanced tradeoffs in which no objective can be improved without degrading another [20]. However, existing MORL approaches have not yet been effectively adapted to packet routing in satellite networks, leaving an important gap to be addressed.

To address these unresolved challenges, there is a clear need for a packet routing framework that can: 1) jointly optimize multiple, often conflicting, performance objectives; 2) explicitly incorporate congestion awareness into decision-making; and 3) maintain scalability under highly dynamic LEO topologies. Motivated by this gap, we propose pareto-optimal multi-agent proximal policy optimization packet routing framework (POMAP), a Pareto-optimal multiobjective optimization framework for packet routing in satellite networks. Unlike conventional weighted-sum RL approaches, POMAP dynamically balances key performance metrics without requiring heuristic weight assignment, thereby avoiding bias and improving adaptability in realistic satellite environments. Within POMAP, we design a packet routing method for large-scale LEO satellite networks that integrates multiagent proximal policy optimization (MAPPO) with graph attention networks (GATs). MAPPO enables decentralized learning of efficient packet routing policies, while GATs capture spatial dependencies and model dynamic satellite interactions, thereby allowing agents to prioritize critical links and adapt more effectively to rapidly changing topologies. To the best of authors' knowledge, this is the first work to incorporate multiobjective MARL into packet routing for satellite networks. Furthermore, a systematic comparison with existing methods is presented in Table I, highlighting the distinct advantages of the proposed framework. Based on these insights, the main contributions of this article are summarized as follows.

1) *Dynamic Multiattribute Graph Representation:* We design a dynamic graph representation for LEO satellite networks that captures temporal variations by discretizing system time and embedding multiple edge attributes, which provides a richer and more realistic modeling foundation compared with conventional static graph models.
2) *Queue-Aware Packet Routing Mechanism:* We introduce a queue-aware routing mechanism by modeling each node as a G/G/1/K queue with active queue management (AQM). Unlike prior link-state-based methods, this design explicitly incorporates node-level congestion and queueing delay as key optimization objectives.
3) *Pareto-Optimal Multiobjective Packet Routing Framework:* We develop a Pareto-based multiobjective routing framework that integrates MAPPO with GATs to jointly optimize delay, energy, and reliability. This approach enhances the ability of agents to process graph-structured observations and adapt to dynamic topologies, avoiding the bias of conventional weighted-sum RL methods.
4) *Comprehensive Performance Evaluation:* We conduct extensive experiments on realistic LEO satellite constellations, demonstrating that the proposed POMAP framework consistently outperforms state-of-the-art baselines in terms of cumulative reward, Pareto front coverage, delay, energy consumption, and packet loss rate.

The remainder of this article is organized as follows. Section II reviews related work on packet routing in satellite networks. Section III introduces the system model, and Section IV formulates the routing optimization problem. Section V presents the proposed method in detail. Section VI describes the simulation setup and presents performance evaluation results. Finally, Section VII concludes this article and discusses potential directions for future research.

II. Related Work

A. RL- and DRL-Based Centralized Packet Routing

Early studies explored RL methods such as Q-routing to enable adaptive packet forwarding in dynamic networks [21], [32]. While these approaches demonstrated the potential of RL for routing, they suffered from scalability limitations due to the curse of dimensionality, where the exponential growth of the state space restricted their applicability and hindered the effective exploitation of historical traffic patterns for route selection.

The emergence of deep RL (DRL) mitigated some of these limitations by employing deep neural networks to process high-dimensional inputs and learn complex routing policies [22]. For example, Liu et al. [22] applied an approach based on deep Q-network (DQN) to minimize end-to-end delays in aeronautical ad hoc networks. However, centralized DRL generally assumes global observations or relies on a single agent, which leads to substantial computational and communication overhead, susceptibility to single point of failure, and inadequate modeling of internode interactions. Consequently, such centralized strategies are impractical for large-scale dynamic networks and naturally motivate the adoption of decentralized solutions, particularly in satellite networks.

B. MARL- and Queueing-Based Decentralized Packet Routing

Unlike centralized schemes, MARL enables nodes to make routing decisions independently while coordinating with others, thereby utilizing local information and improving robustness. You et al. [23] proposed deep Q-routing with communication, a fully distributed framework that enhances resource utilization and packet delivery, while Chen et al. [12] introduced a MAPPO-based meta-learning mechanism to adapt policies across diverse traffic patterns. Similar MARL approaches have been investigated in integrated access/backhaul, tactical sensor networks, uncrewed aerial vehicle swarms, and underwater networks [33], [34], [35], [36]. For satellite networks, Huang et al. [24] proposed a packet routing method to alleviate centralized inefficiencies and reduce delay, and Xu et al. [13] designed a spatially assisted MARL scheme that jointly optimizes queueing and propagation delays.

Complementary to MARL, queueing theory provides analytical tools for modeling congestion dynamics and delay metrics. Early studies employed queueing models for scheduling and tandem analysis [37], [38], and later examined convergence between discrete packet routing and continuous flow models [39]. More recent works have integrated queueing with RL/DRL for dynamic routing [40], [41], [42] although these approaches are often centralized or based on independent learning. Building on this foundation, researchers combined MARL with queueing theory to improve performance in diverse networks, including service differentiation, delay prediction, and quality of service (QoS)-oriented strategies [25], [27], [28], as well as flow-centric routing for LEO satellites [26]. However, existing solutions continue to encounter scalability issues and lack effective multiobjective optimization for dynamic satellite networks.

Despite these advances, most MARL-queueing solutions still rely on DQN-like algorithms with discrete action spaces and extensive offline tuning, which limit scalability in dynamic settings. Moreover, their reliance on weighted-sum objectives (e.g., minimizing latency while maximizing throughput) can bias results and fail to capture true tradeoffs. These limitations highlight the need for MORL frameworks, particularly Pareto-based methods, to achieve more balanced routing decisions.

C. Multiobjective Optimization-Based Packet Routing

Traditional RL/DRL routing typically combines multiple metrics (latency, energy, and throughput) into a weighted sum, where small weight changes can drastically alter routing behavior and complicate the process of weight selection [43]. Such methods often exhibit rigid tradeoffs and limited adaptability in dynamic networks. For example, Wei et al. [29] introduce a Pareto dominance-based routing algorithm that considers delay, congestion, and bandwidth, but it remains centralized and lacks scalability. Similarly, Lyu et al. [15] develop an MARL-based scheme that minimizes delay while ensuring energy efficiency and packet loss constraints, yet it still relies on weight-based optimization.

Several studies have applied Pareto-based optimization to routing in different domains. For instance, Zheng et al. [44] apply a DRL approach for path planning, evaluating tradeoffs using hypervolume metrics and nondominated solution (NDS) sets; Ghorai et al. [30] formulate routing in the Internet of Vehicles as a multiobjective problem using an evolutionary algorithm to optimize delay, distance, bandwidth, and connectivity; and Han et al. [31] propose a multiattribute routing method for opportunistic networks that enhances node centrality and similarity while reducing energy and cache usage. However, these Pareto-based methods lack RL-driven adaptability in highly dynamic network environments. This limitation motivates the integration of Pareto-based optimization with RL, namely, MORL.

MORL has emerged as an effective approach for multiobjective optimization. Unlike traditional RL that aggregates objectives into a single scalar reward, MORL maintains separate reward functions to facilitate tradeoff exploration. Pareto-based MORL further identifies Pareto-optimal solutions, thereby enhancing interpretability in complex network scenarios [45]. Building on these insights, we propose a MAPPO-based packet routing framework that incorporates GATs and queueing models, effectively addressing the limitations of existing multiobjective approaches in dynamic satellite networks. Details of the proposed method are presented in Section V.

III. System Model

A. Network Model

To accurately model the dynamic topology of LEO satellite networks, the system is represented as a sequence of time-evolving graphs over discrete time intervals. System time is divided into discrete slots, denoted as $t \in \mathcal{T} = \{1, 2, \ldots, T\}$. At each time slot t, the satellite network is modeled as a static graph $G(t) = (V, E(t))$, where V denotes the set of satellite nodes and $E(t)$ represents the set of intersatellite links (ISLs) active at time t. This study focuses on packet routing within a single layer of the satellite constellation, and thus, the network model is based on a single-shell configuration. The ISL configuration follows the Grid+ model [46], in which each satellite establishes two intraplane links with neighbors in the same orbital plane and two interplane links with satellites in adjacent planes. As defined by this configuration, each satellite maintains up to four ISLs throughout its operation. The key notations used in this article are summarized in the Nomenclature.

B. Communication Model

1) Transmission Delay: We first consider the transmission delay over free-space optical (FSO) ISLs. Following ITU-R P.525-5 [47], the FSPL of link $e(i, j)$ is expressed as:

$$L_{ij}(t) = \left(\frac{4\pi d_{ij}(t) f}{c}\right)^2 \tag{1}$$

where f is the carrier frequency, $d_{ij}(t)$ is the distance between satellites i and j, and c is the speed of light.

The system assumes an ideal, symmetric free-space channel, where the SNR between satellites i and j is expressed as

$$\mathrm{SNR}_{ij}(t) = \frac{\left|h_{ij}(t)\right|^2 P_{\mathrm{tx}}}{L_{ij}(t)\, N_{\mathrm{total}}} \tag{2}$$

where P_{tx} denotes the transmission power, $L_{ij}(t)$ represents the FSPL, and $N_{\mathrm{total}} = N_0 B$ is the total noise power, with N_0 denoting the noise spectral density and B denoting the bandwidth. The factor $|h_{ij}(t)|$ captures small-scale fading (set to 1 for an ideal link), so attenuation is entirely characterized by $L_{ij}(t)$.

According to Shannon's theorem, the maximum data rate between satellites i and j can be expressed as

$$R_{ij}(t) = B \log_2\left(1 + \mathrm{SNR}_{ij}(t)\right) \tag{3}$$

where B (Hz) denotes the channel bandwidth.

The transmission delay depends on both the amount of data scheduled for transmission and the instantaneous link capacity. Let $s_{ij}(t)$ represent the number of packets scheduled for transmission from node i to node j during time slot t, as determined by the scheduling mechanism introduced in the subsequent queueing model. Assuming serialized transmission, the delay over ISL $e(i, j)$ is given by

$$D_{ij}^{\mathrm{trans}}(t) = \frac{s_{ij}(t) \cdot S}{R_{ij}(t)} \tag{4}$$

where S is the packet size in bits.

2) Propagation Delay: Satellite nodes communicate through intersatellite optical links, and the propagation delay between satellites i and j is expressed as

$$D_{ij}^{\mathrm{prop}}(t) = \frac{d_{ij}(t)}{c} \tag{5}$$

where $d_{ij}(t)$ denotes the distance between satellites i and j at time t, and c represents the speed of light.

3) Queueing Delay: Each satellite node i maintains M queues, indexed by $m \in \mathcal{M} = \{1, 2, \ldots, M\}$, each corresponding to a specific traffic class or flow, with arrivals following a general distribution consistent with the $G/G/1/K$ model. To support heterogeneous traffic requirements, each node employs a WPQ strategy. WPQ adopts a strict priority discipline across predefined priority levels; queues in higher levels always preempt those in lower levels, while within the same level, service is allocated proportionally to the assigned weights. This combines first-in first-out (FIFO) order inside each queue with priority-based allocation across queues.

a) Node-level service capacity constraint: The total forwarding capacity of node i during time slot t is constrained by the physical transmission rates of its active ISLs. Specifically, the node-level service rate $\mu_i(t)$, expressed in packets/s, is defined as

$$\mu_i(t) = \sum_{j \in N_i(t)} \frac{R_{ij}(t)}{S} \tag{6}$$

where $R_{ij}(t)$ denotes the transmission rate from node i to neighbor j (in bits/s), $N_i(t)$ is the set of neighbors of node i at time t, and S is the packet size in bits. The effective service capacity within a slot of duration Δt is, then, $\mu_i(t)\Delta t$, which provides the node-level service bound.

b) WPQ model: Each queue m at node i is assigned a time-varying weight $w_{i,m}(t)$ during slot t. In the WPQ model, service rates are first allocated across priority levels according to strict priority: queues with higher priority levels preempt those with lower levels. Within the same priority level, backlogged queues share the service rate in proportion to their current weights.

Let $B_i(t)$ denote, in slot t, the set of backlogged queues at node i that belong to the highest nonempty priority level. Then, the service rate $\mu_{i,m}(t)$ for queue m is

$$\mu_{i,m}(t) = \begin{cases} \dfrac{w_{i,m}(t)}{\sum_{k \in B_i(t)} w_{i,k}(t)} \cdot \mu_i(t), & m \in B_i(t) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where we assume that $\sum_{k \in B_i(t)} w_{i,k}(t) > 0$ whenever $B_i(t)$ is nonempty. This ensures that $\mu_{i,m}(t) > 0$ for all $m \in B_i(t)$, so that the per-class delay approximation in (10) is always well-defined.

c) Queue length update: In time slot t, the length of queue $q_{i,m}(t)$ is updated as

$$q_{i,m}(t) = \min\left\{\max\left[q_{i,m}(t-1) + r_{i,m}(t-1) - x_{i,m}(t-1), 0\right], K_m\right\} \tag{8}$$

where $r_{i,m}(t-1)$ is the number of class-m arrivals, $x_{i,m}(t-1)$ is the number of packets served, and K_m is the capacity of queue m with $\sum_{m=1}^{M} K_m = K$. This equation ensures finite queue capacity, with overflow packets dropped.

The number of packets served in time slot $t-1$ is computed as

$$x_{i,m}(t-1) = \min\left\{q_{i,m}(t-1), \mu_{i,m}(t-1)\Delta t\right\} \tag{9}$$

where Δt denotes the slot duration. When slot-level scheduling is adopted, any unused service capacity is dynamically redistributed to active queues to improve bandwidth utilization. This formulation ensures that the backlog never exceeds K_m and remains consistent with the service capacity $\mu_{i,m}(t)$.

d) Queueing delay calculation: In each time slot, the per-class queueing delay, denoted as $D_{i,m}^{\text{queue}}(t)$ for traffic class m at node i, is approximated as

$$D_{i,m}^{\text{queue}}(t) \approx \frac{q_{i,m}(t)}{\mu_{i,m}(t)}, \quad m \in B_i(t) \tag{10}$$

where $q_{i,m}(t)$ is the backlog of class m at node i and $\mu_{i,m}(t)$ is its service rate. This equation approximates delay via Little's law and is defined only for queues in the active backlogged set $B_i(t)$, i.e., those belonging to the highest nonempty priority level. For empty queues, the delay is naturally zero. For queues outside $B_i(t)$ that are backlogged but receive no service when higher priority queues are nonempty, the queueing delay is not defined within this approximation since they experience starvation until the higher priority queues are cleared.

The aggregate queueing delay at node i, denoted as $D_i^{\text{queue}}(t)$, is defined as the maximum delay among all backlogged queues. To ensure well-defined behavior even when no queues are backlogged, we adopt the convention that the aggregate delay is zero in this case

$$D_i^{\text{queue}}(t) = \begin{cases} \max\limits_{m \in B_i(t)} D_{i,m}^{\text{queue}}(t), & \text{if} \quad B_i(t) \neq \emptyset \\ 0, & \text{if} \quad B_i(t) = \emptyset. \end{cases} \tag{11}$$

By this definition, $D_i^{\text{queue}}(t)$ consistently reflects the maximum per-class queueing delay when queues are active, and is zero when the node is completely idle. Note that, since $B_i(t)$ is defined as the set of backlogged queues within the highest nonempty priority level, the aggregate delay corresponds to the worst-case latency of the currently dominant priority class, while lower priority queues are excluded because they receive no service whenever higher priority queues are backlogged.

e) Active queue management: To proactively manage congestion and enhance buffer utilization, each queue m at node i employs an AQM mechanism. This mechanism dynamically adjusts the packet drop probability based on the current queue length. Under WPQ, congestion control for high-priority queues is critical, as their backlog directly impacts delay-sensitive services.

At the beginning of each time slot t, node i selects AQM parameters $\alpha_i(t), \beta_i(t)$, and $p_{\max,i}(t)$ as part of its control action (see Section V-B). These parameters are applied uniformly across all queues at the node to reduce implementation overhead and ensure consistent AQM behavior. The thresholds for queue m are, therefore, defined as fractions of the maximum queue length K_m

$$\delta_{1,m}(t) = \alpha_i(t) K_m, \delta_{2,m}(t) = \beta_i(t) K_m$$
$$0 \le \alpha_i(t) < \beta_i(t) \le 1. \tag{12}$$

A queue is considered underutilized if its length is below $\delta_{1,m}(t)$ and overloaded if it exceeds $\delta_{2,m}(t)$. The AQM mechanism adjusts the packet drop probability $p_m(q_{i,m}(t))$ according to the following rule:

$$p_m(q_{i,m}(t)) = \begin{cases} 0, & q_{i,m}(t) < \delta_{1,m}(t) \\ p_{\max,i}(t)\left(\dfrac{q_{i,m}(t) - \delta_{1,m}(t)}{\delta_{2,m}(t) - \delta_{1,m}(t)}\right)^2, & \delta_{1,m}(t) \le q_{i,m}(t) < \delta_{2,m}(t) \\ p_{\max,i}(t), & q_{i,m}(t) \ge \delta_{2,m}(t). \end{cases} \tag{13}$$

The quadratic segment provides a smooth increase of the drop probability between the two thresholds; when the queue length exceeds the upper threshold, the probability stays at the plateau $p_{\max,i}(t)$, ensuring continuity at $q = \delta_{2,m}(t)$.

For clarity, the unified parameters across all queues at node i are $\alpha_i(t)$,$\beta_i(t)$, and $p_{\max,i}(t)$, while the resulting thresholds remain queue-specific

$$\forall m \in \mathcal{M}: \quad \begin{aligned} &\delta_{1,m}(t) = \alpha_i(t)\, K_m, \\ &\delta_{2,m}(t) = \beta_i(t)\, K_m, \quad 0 \le \alpha_i(t) < \beta_i(t) \le 1 \\ &p_{\max,m}(t) = p_{\max,i}(t). \end{aligned} \tag{14}$$

This design guarantees that all traffic classes at a node employ the same AQM parameters in each time slot, while allowing these parameters to adapt dynamically over time. Under WPQ scheduling, packet drops in high-priority queues exert a greater influence on end-to-end performance; consequently, the AQM mechanism effectively enforces priority-aware congestion control.

f) Notification and packet loss: When the queue length $q_{i,m}(t)$ reaches its maximum capacity K_m, the buffer for traffic class m becomes full, and any newly arriving packets of this class are dropped immediately. To mitigate congestion and prevent sustained packet loss, the node generates a congestion notification and transmits it to neighboring satellites via control-plane signaling.

The notification message includes the traffic class index m, the current queue length $q_{i,m}(t)$, and the associated packet drop probability $p_m(q_{i,m}(t))$. Upon reception, neighboring nodes can adjust their routing policies or reduce the forwarding rate toward the congested node. Notifications from high-priority queues are assigned greater urgency due to their scheduling dominance under WPQ. This distributed feedback mechanism facilitates cooperative congestion control, mitigates localized congestion, and improves overall network stability and data flow efficiency.

4) Communication Delay Calculation: Based on the preceding analysis, the total communication delay $D_{ij}(t)$ is the sum of transmission, propagation, and queueing delays, and is defined as

$$D_{ij}(t) = D_{ij}^{\text{trans}}(t) + D_{ij}^{\text{prop}}(t) + D_{i}^{\text{queue}}(t). \tag{15}$$

This aggregate delay metric provides a comprehensive measure of per-link delay performance and serves as the foundation for evaluating routing and network control strategies. Since WPQ scheduling is employed, the queueing delay component $D_i^{\text{queue}}(t)$ primarily reflects the waiting time determined by queue priorities, where higher priority classes experience shorter delays compared to lower priority ones.

C. Energy Model

The energy model focuses on the consumption of satellite nodes during packet forwarding. The total energy consumption $E_i(t)$ of a satellite node in slot t includes the energy required for both data transmission and reception. The presence and utilization of an ISL, indicated by $a_{ij}(t)$, directly determine the node's energy cost.

1) Transmission Energy: The energy consumed by satellite node i for transmitting data packets in time slot t is defined as

$$E_i^{\text{tx}}(t) = P_{\text{tx}} \cdot \sum_{j \in N_i(t)} a_{ij}(t)\, \frac{S \cdot s_{ij}(t)}{R_{ij}(t)} \tag{16}$$

where P_{tx} denotes the transmit power (W), $s_{ij}(t)$ denotes the number of packets sent from i to neighbor j, S is the packet size (bits), and $R_{ij}(t)$ is the transmission rate (bit/s). The binary variable $a_{ij}(t) \in \{0,1\}$ indicates the availability of the ISL between nodes i and j. The term $S \cdot s_{ij}(t)$ corresponds to the total transmitted bits, and dividing by $R_{ij}(t)$ yields the transmission duration. Multiplying by P_{tx} produces the transmission energy (J). When $a_{ij}(t) = 0$, the link is unavailable, and no transmission energy is consumed.

2) Reception Energy: The energy consumed by node i for receiving data in time slot t is defined as

$$E_i^{\text{rx}}(t) = P_{\text{rx}} \cdot \sum_{j \in N_i(t)} a_{ji}(t)\, \frac{S \cdot u_{ji}(t)}{R_{ji}(t)} \tag{17}$$

where P_{rx} denotes the receive power (W), $u_{ji}(t)$ denotes the number of packets received from neighbor j, and $R_{ji}(t)$ is the corresponding transmission rate. Analogous to the transmission case, this expression represents the reception time multiplied by the receive power, yielding the total reception energy. The factor $a_{ji}(t)$ ensures that reception energy is only consumed when the incoming ISL from j to i is available.

3) Total and Remaining Energy: The overall energy consumption of satellite node i in time slot t is defined as the sum of transmission and reception energies

$$E_i(t) = E_i^{\text{tx}}(t) + E_i^{\text{rx}}(t). \tag{18}$$

The remaining energy of node i at time t is calculated as

$$E_i^{\text{rem}}(t) = E_i^{\text{init}} - \sum_{\tau=1}^{t} E_i(\tau) \tag{19}$$

where E_i^{init} denotes the initial energy allocated to the satellite, and $E_i(\tau)$ represents the total energy consumed in slot τ. This cumulative formulation captures the gradual depletion of onboard energy resources, thereby constraining the long-term sustainability of routing strategies.

IV. Problem Formulation

A. Pareto Optimization Preliminary

Many real-world optimization problems involve multiple conflicting objectives that must be optimized concurrently. These problems, referred to as multiobjective optimization problems (MOPs), are formally defined as follows.

Definition 1 (MOP): An MOP is defined as

$$\min_{\mathbf{x} \in X} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})) \tag{20a}$$

$$\text{s.t. } \mathbf{x} \in X \tag{20b}$$

where $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is the vector of decision variables, X represents the feasible solution space defined by constraints, and $\mathbf{f(x)} = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x}))$ is the objective function vector, where each $f_\ell(\mathbf{x})$ is an individual objective to be minimized.

Unlike single-objective optimization problems, which aim to identify a single optimal solution, MOPs typically yield a set of tradeoff solutions, where improving one objective may result in the degradation of another. This introduces the concept of Pareto optimality, which serves as a fundamental principle in multiobjective optimization.

Definition 2 (Pareto Dominance): A solution $\mathbf{x}_1 \in X$ Pareto dominates another solution $\mathbf{x}_2 \in X$, denoted as $\mathbf{x}_1 \prec \mathbf{x}_2$, if

$$\forall \ell \in \{1, \ldots, m\}, \quad f_\ell(\mathbf{x}_1) \le f_\ell(\mathbf{x}_2) \tag{21}$$

and at least one strict inequality holds

$$\exists \ell' \in \{1, \ldots, m\}, \quad f_{\ell'}(\mathbf{x}_1) < f_{\ell'}(\mathbf{x}_2). \tag{22}$$

This implies that $\mathbf{x}_1$ is strictly better than $\mathbf{x}_2$ in at least one objective while being no worse in any other objective.

Definition 3 (Pareto-Optimal Solution): A solution $\mathbf{x}^* \in X$ is a Pareto-optimal solution if no other solution $\mathbf{x} \in X$ Pareto dominates $\mathbf{x}^*$, i.e.,

$$\nexists \mathbf{x} \in X, \quad \mathbf{f}(\mathbf{x}) \prec \mathbf{f}(\mathbf{x}^*). \tag{23}$$

This implies that improving any objective of $\mathbf{x}^*$ would necessarily degrade at least one other objective.

Definition 4 (Pareto Front): The set of all Pareto-optimal solutions forms the Pareto front, defined as

$$\mathrm{PF} = \left\{ \mathbf{f}(\mathbf{x}) \,\middle|\, \mathbf{x} \in X, \nexists \mathbf{x}' \in X, \mathbf{f}\left(\mathbf{x}'\right) \prec \mathbf{f}(\mathbf{x}) \right\}. \tag{24}$$

The Pareto front represents the tradeoff surface where no objective can be improved without degrading another.

Pareto optimization offers a foundational framework for addressing MOPs by identifying a set of tradeoff solutions rather than a single global optimum. The notions of Pareto dominance, Pareto optimality, and the Pareto front enable decision-makers to assess feasible solutions and select tradeoffs aligned with their preferences.

B. MOP Formulation

In satellite networks, communication delay, energy consumption, and packet loss rate are three critical yet conflicting performance metrics. For example, reducing delay often requires higher transmission power and thus increases energy consumption, while minimizing packet loss may introduce longer queueing delays through AQM. Conventional single-objective approaches that combine these metrics into a weighted sum are generally insufficient in dynamic network environments. To balance these tradeoffs under highly dynamic topology, limited resources, and latency-sensitive services, we adopt a Pareto optimization framework, which identifies routing strategies where improving one objective necessarily involves compromise in others. Based on this principle, we formulate an MOP that simultaneously considers the three metrics, as detailed in the following:

$$\min_{\mathbf{x} \in X} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), f_3(\mathbf{x})) \tag{25a}$$

$$f_1(\mathbf{x}) = \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}(t)} x_p(t) \sum_{(i,j) \in p} D_{ij}(t) \tag{25b}$$

$$f_2(\mathbf{x}) = \sum_{t \in \mathcal{T}} \sum_{i \in V} E_i(t) \tag{25c}$$

$$f_3(\mathbf{x}) = \sum_{t \in \mathcal{T}} \sum_{i \in V} \sum_{m \in \mathcal{M}} p_m\left(q_{i,m}(t)\right) \tag{25d}$$

$$\sum_{(i,j) \in p} D_{ij}(t) \le D_{\max}$$

$$\forall p \in \mathcal{P}(t) \text{ with } x_p(t) > 0, \; t \in \mathcal{T} \tag{25e}$$

$$\sum_{\tau=1}^{t} E_i(\tau) \le E_i^{\text{init}} \; \forall i \in V, \; t \in \mathcal{T} \tag{25f}$$

$$\sum_{m \in \mathcal{M}} q_{i,m}(t) \le K \; \forall i \in V, \; t \in \mathcal{T}. \tag{25g}$$

The objective function includes three components: end-to-end delay [see (25b)], total energy consumption [see (25c)], and congestion-induced packet loss [see (25d)]. The delay term accounts for transmission, propagation, and queueing delays along the selected paths, with WPQ ensuring that higher priority traffic experiences shorter waiting times. The energy term aggregates transmission and reception costs across all nodes, reflecting the sustainability of routing. The packet loss term is expressed via drop probabilities from buffer overflows and AQM, which serves as a tractable surrogate for actual packet losses. Together, these three objectives capture the fundamental tradeoffs among latency, energy efficiency, and reliability in LEO satellite routing.

The feasible region is defined by three classes of constraints. First, the end-to-end delay of any selected path must not exceed $D_{\max}$ [see (25e)], which ensures latency compliance for delay-sensitive services. Second, the cumulative energy consumed by each satellite up to time t cannot exceed its initial budget E_i^{init} [see (25f)], ensuring that residual energy remains nonnegative. Third, the buffer occupancy of each node must remain within its maximum capacity K [see (25g)], thereby preventing unbounded queue growth. Collectively, these constraints ensure that the optimization problem remains physically realizable and that routing strategies satisfy the critical requirements of delay, energy, and congestion control.

V. Proposed Method

A. Decomposition-Based Pareto Optimization Method

MOPs frequently arise in satellite communication scenarios, where conflicting objectives such as delay minimization, energy efficiency, and congestion control must be jointly addressed. A widely adopted strategy for solving MOPs is the decomposition-based method, which transforms the original problem into multiple scalar optimization subproblems [48], [49]. Each subproblem represents a specific tradeoff preference and, when considered collectively, approximates the Pareto front. The decomposition approach employs scalarization techniques such as the weighted sum, Chebyshev distance, and penalty-based boundary intersection. In this study, we adopt the weighted sum method due to its simplicity and low computational cost.

Formally, let $\lambda^1, \lambda^2, \ldots, \lambda^J$ denote a set of uniformly distributed weight vectors, where each $\lambda^j = (\lambda_1^j, \lambda_2^j, \ldots, \lambda_m^j)^\top$ represents the relative importance of the m objective functions. The jth subproblem is formulated as

$$\min_{\mathbf{x} \in X} g^{\text{ws}}\left(\mathbf{x} \,\middle|\, \lambda^j\right) = \sum_{\ell=1}^{m} \lambda_\ell^j f_\ell(\mathbf{x}) \tag{26}$$

where $\mathbf{x} \in X$ denotes the decision variable within the feasible solution space, and $f_\ell(\mathbf{x})$ is the ℓth objective function. Each weight vector satisfies $\lambda_\ell^j \ge 0$ and $\sum_{\ell=1}^{m} \lambda_\ell^j = 1$, ensuring a valid convex combination of objectives. Solving each subproblem produces a Pareto-optimal solution corresponding to a specific tradeoff vector, and the collection of these solutions collectively approximates the complete Pareto front.

To enhance training efficiency, we introduce a neighborhood-based parameter transfer mechanism: since adjacent subproblems with similar weight vectors have nearby optimal solutions, the optimized parameters $[\theta^*_{\lambda^{j-1}}, \phi^*_{\lambda^{j-1}}]$ of subproblem $(j-1)$ are used to initialize $[\theta_{\lambda^j}, \phi_{\lambda^j}]$ for subproblem j, thereby accelerating convergence. The overall procedure of this decomposition-based Pareto optimization with neighborhood-based transfer learning is illustrated in Fig. 1. We further incorporate this decomposition strategy into an MARL framework to address partial observability and decentralized control inherent in satellite networks. Each subproblem is optimized using an MARL-based method, enabling agents to learn cooperative policies that effectively balance competing objectives. The integration of the decomposition strategy into the MARL framework is illustrated in Fig. 2. This approach results in the formulation of a POMDP, which specifies the observation space, action space, and scalarized reward function tailored to each weight vector, as detailed in Section V-B.

B. POMDP Design

In MARL scenarios, agents typically operate under partial observability, relying on local observations to make decisions rather than having access to the full global state. This setting can be effectively modeled using a POMDP, which is defined as

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, Z, R, \gamma) \tag{27}$$

where $\mathcal{S}$ is the set of latent environmental states, $\mathcal{A}$ denotes the joint action space of all agents, and $P(s' \mid s, \mathbf{a})$ represents the state transition probability given joint action $\mathbf{a}$. The reward function $R(s, \mathbf{a})$ evaluates the system performance based on criteria such as latency, energy efficiency, or congestion. Each agent receives local observations $\mathbf{o}$ from the observation function $Z(\mathbf{o} \mid s', \mathbf{a})$, and $\gamma \in [0, 1)$ is the discount factor that controls the tradeoff between immediate and future rewards.

Since the decomposition-based Pareto optimization approach reformulates the original multiobjective routing problem into a set of scalar subproblems, each subproblem can be addressed within the standard POMDP framework. Rather than handling multiple conflicting objectives simultaneously, each agent optimizes a scalar reward function corresponding to a specific tradeoff among key routing metrics such as delay, energy consumption, and packet loss rate. With this POMDP structure in place, the design of states, actions, and rewards is specified to enable effective packet routing policies under partial observability, and the details are presented as follows.

1) Observation Space: The observation space is designed to capture local network conditions and traffic dynamics, enabling each satellite node to make informed routing decisions. For node i at time t, it is defined as

$$O_i(t) = \left[\, Q_i(t),\, \Delta_i(t),\, E_i^{\text{rem}}(t),\, \eta_i(t) \,\right]. \tag{28}$$

Here, the elements are defined as follows.

1) *Queue Status $Q_i(t)$:* Queue length $q_{i,m}(t)$ and packet drop probability $p_m(q_{i,m}(t))$ for each traffic class m
$$Q_i(t) = \left\{ q_{i,m}(t),\, p_m\left(q_{i,m}(t)\right) \right\}_{m=1}^{M}. \tag{29}$$
2) *Link Delays $\Delta_i(t)$:* Per-link delays from i to each neighbor j
$$\Delta_i(t) = \left\{ D_{ij}(t) \right\}_{j \in N_i(t)}. \tag{30}$$
3) *Remaining Energy $E_i^{\text{rem}}(t)$:* The remaining energy of node i, as defined in (19).
4) *Neighbor Information $\eta_i(t)$:* The number of neighbors $|N_i(t)|$ and connectivity indicators $a_{ij}(t)$
$$\eta_i(t) = \left(|N_i(t)|, \left\{a_{ij}(t)\right\}_{j \in N_i(t)}\right). \tag{31}$$

This representation jointly reflects queue status, link conditions, energy constraints, and topology, providing sufficient information for satellites to optimize routing while balancing efficiency and fairness.

2) Action Space: In our model, each satellite node i makes decisions regarding packet routing, queue scheduling, and AQM to optimize network performance, with a focus on latency and energy consumption. The action space for each agent comprises three key components: traffic distribution, WPQ weight assignment, and AQM parameter adjustment. It is defined as

$$\boldsymbol{a}_i(t) = \Bigg[\underbrace{\left(\rho_{ij}(t)\right)_{j \in N_i(t)}}_{\text{Traffic Distribution}},\ \underbrace{\left(w_{i,1}(t), \ldots, w_{i,M}(t)\right)}_{\text{WPQ Weights}} \underbrace{\left(\alpha_i(t), \beta_i(t), p_{\max,i}(t)\right)}_{\text{AQM Parameters}} \Bigg]. \tag{32}$$

Here, the elements are defined as follows.

1) *Traffic Distribution $\boldsymbol{\rho}_i(t)$:* The traffic distribution vector $\boldsymbol{\rho}_i(t) = (\rho_{ij}(t))_{j \in N_i(t)}$ specifies the proportion of packets that node i forwards to each neighbor j at time t, subject to
$$\sum_{j \in N_i(t)} \rho_{ij}(t) = 1, \rho_{ij}(t) \geq 0 \ \forall j \in N_i(t). \tag{33}$$
2) *WPQ Weights $[w_{i,1}(t), \ldots, w_{i,M}(t)]$:* These values represent the relative service weights of the M traffic classes at node i. A higher $w_{i,m}(t)$ indicates that class-m packets receive a larger share of service among queues of the same priority level, consistent with the WPQ discipline.
3) *AQM Parameters $[\alpha_i(t), \beta_i(t),$ and $p_{\max,i}(t)]$:* These parameters define the queue-length thresholds and the packet drop probability of the AQM mechanism at node i. Specifically, for each queue m, the thresholds are defined as
$$\delta_{1,m}(t) = \alpha_i(t) K_m, \delta_{2,m}(t) = \beta_i(t) K_m \tag{34}$$
with $0 \leq \alpha_i(t) < \beta_i(t) \leq 1$, and the maximum drop probability is $p_{\max,i}(t)$. All queues at node i share the same $\alpha_i(t), \beta_i(t)$, and $p_{\max,i}(t)$ to ensure consistent AQM configuration.

This action space enables dynamic traffic control, facilitating efficient routing, priority-based queue scheduling, and adaptive congestion management.

3) Reward Function: To achieve Pareto optimization of delay, energy consumption, and packet loss, the reward function is designed using the weighted-sum scalarization approach. As defined in (26), each subproblem is formulated as a weighted sum of multiple objectives based on a specific weight vector λ^j.

Since the underlying objectives are costs to be minimized, we first normalize each cost component to the range $[0, 1]$ using min–max normalization. Let

$$\mathbf{c}_t = \left(c_t^{\text{delay}}, c_t^{\text{energy}}, c_t^{\text{loss}}\right) \tag{35}$$

denote the normalized cost vector at time step t. The scalarized reward is, then, defined as the negative weighted sum of normalized costs

$$\tilde{r}_t = -g^{\text{ws}}\left(\mathbf{c}_t \mid \lambda^j\right) = -\sum_{\ell=1}^{m} \lambda_\ell^j c_t^{(\ell)}. \tag{36}$$

Here, $\lambda^j = (\lambda_1^j, \lambda_2^j, \lambda_3^j)^\top$ represents the tradeoff preferences for the jth subproblem, where λ_1^j, λ_2^j, and λ_3^j denote the relative importance of delay, energy consumption, and packet loss, respectively. This formulation ensures that maximizing cumulative reward in RL is equivalent to minimizing the original multiobjective costs, while allowing different subproblems to explore distinct tradeoffs.

It should be emphasized that this normalization is employed solely for reward shaping in RL to stabilize training. The underlying MOP in Section IV remains defined in terms of the original physical metrics (delay in s, energy in J, and packet-loss probability). This decomposition-based reward design not only guarantees consistency between RL optimization and the underlying cost-minimization problem, but also allows the learned policies across all subproblems to approximate the Pareto front collectively. Building on this design, our proposed method employs MAPPO to optimize the formulated POMDP framework, as illustrated in Fig. 3, with the detailed algorithm presented in Section V-E.

C. MAPPO Preliminary

MAPPO extends the standard PPO algorithm to multiagent settings under the centralized training with decentralized execution (CTDE) paradigm. This makes it well-suited for multiagent POMDP environments such as satellite routing, where agents act on partial observations while benefiting from a centralized critic during training.

To efficiently address multiobjective optimization under partial observability, MAPPO is combined with a decomposition-based Pareto optimization approach. Instead of optimizing a single multiobjective policy, the original problem is decomposed into multiple scalar subproblems, each defined by a distinct weight vector λ^j that encodes a specific tradeoff among objectives. Each subproblem is optimized independently using MAPPO, and the resulting policies collectively approximate the Pareto front. Furthermore, a neighborhood-based parameter transfer strategy accelerates training by leveraging the similarity between parameter sets of adjacent subproblems.

Each agent's policy is optimized using the clipped surrogate PPO objective

$$L^{\text{CLIP}}(\theta_i) = \mathbb{E}\left[\min\left(\rho_t^i(\theta_i) A_t^i,\ \text{clip}\left(\rho_t^i(\theta_i), 1-\epsilon, 1+\epsilon\right) A_t^i\right)\right] \tag{37}$$

where $\rho_t^i(\theta_i)$ is the importance sampling ratio

$$\rho_t^i(\theta_i) = \frac{\pi_{\theta_i}\left(a_t^i \middle| o_t^i\right)}{\pi_{\theta_{i,\text{old}}}\left(a_t^i \middle| o_t^i\right)}. \tag{38}$$

The centralized critic V_ϕ estimates the global state value by minimizing the loss

$$L_V(\phi) = \mathbb{E}\left[\left(V_\phi(s_t) - \hat{V}_t\right)^2\right] \tag{39}$$

where the one-step bootstrapped target value is defined as

$$\hat{V}_t = \tilde{r}_t + \gamma V_\phi\left(s_{t+1}\right). \tag{40}$$

The advantage function is estimated using generalized advantage estimation (GAE)

$$A_t^i = \sum_{k=0}^{\infty} \left(\gamma\lambda_{\text{GAE}}\right)^k \delta_{t+k} \tag{41}$$

where λ_{GAE} is the GAE parameter. The temporal difference (TD) residual is defined using the scalarized reward $\tilde{r}_t$

$$\delta_t = \tilde{r}_t + \gamma V_\phi\left(s_{t+1}\right) - V_\phi(s_t) \tag{42}$$

where the scalarized reward $\tilde{r}_t$ is computed by (36).

Each agent's policy parameters are updated via stochastic gradient ascent

$$\theta_i^{k+1} = \theta_i^k + \alpha \nabla_{\theta_i} L^{\text{CLIP}}(\theta_i) \tag{43}$$

where α denotes the learning rate.

Instead of solving a single multiobjective problem, MAPPO optimizes multiple scalarized subproblems derived through the decomposition method. Each subproblem is associated with a weight vector λ^j, which defines the relative importance of objectives. The policy optimization for each subproblem follows the standard MAPPO procedure, and the trained parameters from one subproblem are used to initialize those of neighboring subproblems, enhancing sample efficiency and accelerating convergence. By integrating decomposition-based Pareto optimization with MAPPO, we address a set of scalarized subproblems, each of which is trained independently and warm-started from its neighbors to improve sample efficiency. This approach ensures that each policy is optimized for a specific tradeoff while leveraging information from adjacent subproblems.

D. GAT-Based Neighborhood Encoder

The GAT [50] is employed as a lightweight neighborhood encoder that assigns learnable attention weights to neighboring nodes and encodes each agent's one-hop neighborhood within the time-varying intersatellite graph into a compact representation. This encoder integrates the agent's local observation (as specified in the POMDP design) with features of directly connected neighbors and, when available, edge attributes such as queueing, propagation, or transmission delays as well as link availability. Learned attention coefficients adaptively emphasize informative neighbors, and stacking a small number of layers provides limited multihop awareness while preserving decentralized execution. Multihead attention can also be applied, with heads combined through concatenation or averaging.

Algorithm 1 Decomposition-Based Pareto Optimization With Neighborhood-BasedTransfer Learning

Input: Network topology $G(t)$; initial decentralized actor parameters $\{\theta_i^0\}_{i\in V}$; initial critic parameters ϕ^0; discount factor γ; learning rates; number of subproblems J; ordered weight vectors $\Lambda = \{\lambda^1, \lambda^2, \dots, \lambda^J\}$; MAPPO hyperparameters for Algorithm 2 (e.g., H, $\mathcal{N}$, K_{ppo}, B, ϵ, λ_{GAE}, c_{ent}, c_V, target KL threshold, $R_{\max}$).

Output: Optimized policy map Π^* with entries $\Pi^*[\lambda^j] = \{\pi_{\theta^*_{i,\lambda^j}}\}_{i\in V}$.

Initialize $\Pi^* \leftarrow \emptyset$;
for *each subproblem* $j = 1$ **to** J **do**
    if $j = 1$ **then**
        $\left(\{\theta_{i,\lambda^j}\}_{i\in V},\ \phi_{\lambda^j}\right) \leftarrow \left(\{\theta_i^0\}_{i\in V},\ \phi^0\right)$;
    else
        $\left(\{\theta_{i,\lambda^j}\}_{i\in V},\ \phi_{\lambda^j}\right) \leftarrow \left(\left\{\theta^*_{i,\lambda^{j-1}}\right\}_{i\in V},\ \phi^*_{\lambda^{j-1}}\right)$;
    end
    $\left(\left\{\theta^*_{i,\lambda^j}\right\}_{i\in V},\ \phi^*_{\lambda^j}\right) \leftarrow$ Algorithm 2 $\left(j, \lambda^j, G(t), \{\theta_{i,\lambda^j}\}_{i\in V}, \phi_{\lambda^j}\right)$;
    $\Pi^*\left[\lambda^j\right] \leftarrow \left\{\pi_{\theta^*_{i,\lambda^j}}\right\}_{i\in V}$;
end
return Π^*;

The decentralized actor consumes the GAT embedding instead of the raw observation, while the centralized critic applies a permutation-invariant readout (e.g., mean or attention pooling) over all agents' embeddings to estimate the global value during training. All encoder parameters are optimized jointly with the actor and critic in Algorithm 2, with weight sharing across agents to reduce model size and enhance generalization. Because neighborhood information is captured through the encoder rather than via explicit interagent messaging, no additional online coordination overhead is incurred. The resulting topology- and traffic-aware state representation aligns with the scalarized objective and provides an effective foundation for MAPPO-based policy optimization under partial observability.

E. Proposed MAPPO-Based Packet Routing Method

Algorithm 2 describes the procedure for optimizing the policies of a scalarized subproblem j using MAPPO within a CTDE framework. Each agent i maintains an independent decentralized actor $\pi_{\theta_{i,\lambda^j}}$, while all agents share a single centralized critic $V_{\phi_{\lambda^j}}$ that takes the global state as input for value estimation during training. This optimization procedure is invoked by Algorithm 1 for each decomposed subproblem.

The algorithm proceeds in training rounds for at most $R_{\max}$ rounds. At the beginning of each round, an on-policy buffer $\mathcal{D}^j$ is initialized to store the collected trajectories. For each episode, the environment is reset, and both global and local observations are obtained. At every time step, each agent

Algorithm 2 MAPPO-Based Packet Routing Optimization for Subproblem j

Input: Subproblem index j; weight vector λ^j; topology $G(t)$; initial decentralized policy parameters $\{\theta_{i,\lambda^j}\}_{i\in V}$; initial critic parameters ϕ_{λ^j}; rollout horizon H; episodes per round $\mathcal{N}$; update epochs K_{ppo}; minibatch size B; discount γ; GAE parameter λ_{GAE}; clip ϵ; entropy coeff c_{ent}; value loss coeff c_V; target KL threshold; max training rounds $R_{\max}$.

Output: Updated parameters $\left\{\theta^*_{i,\lambda^j}\right\}_{i\in V}$ and $\phi^*_{\lambda^j}$ for subproblem j.

for *round* $= 1$ **to** $R_{\max}$ **do**
    Initialize on-policy buffer $\mathcal{D}^j \leftarrow \emptyset$;
    for *episode* $= 1$ **to** $\mathcal{N}$ **do**
        Reset env; obtain s_0, $\{o_0^i\}_{i\in V}$;
        for $t = 1$ **to** H **do**
            for *each agent* $i \in V$ **do**
                sample $a_t^i \sim \pi_{\theta_{i,\lambda^j}}(\cdot \mid o_t^i)$;
            end
            Form joint action $\mathbf{a}_t$; step env to get s_{t+1}, $\{o_{t+1}^i\}$, cost vector $\mathbf{c}_t$;
            Set scalarized reward $\tilde{r}_t \leftarrow -g^{\text{ws}}\left(\mathbf{c}_t \mid \lambda^j\right)$ via (36);
            Store $(s_t, \{o_t^i\}, \mathbf{a}_t, \tilde{r}_t, s_{t+1}, \{o_{t+1}^i\})$ in $\mathcal{D}^j$;
        end
    end
    $\{\theta_{i,\text{old},\lambda^j}\} \leftarrow \{\theta_{i,\lambda^j}\}$;
    Compute bootstrapped targets $\hat{V}_t$ via (40) and TD residuals δ_t via (42) using $V_{\phi_{\lambda^j}}(s_t)$;
    Compute advantages A_t^i via GAE (41) (optionally normalize per agent);
    for *epoch* $= 1$ **to** K_{ppo} **do**
        Shuffle $\mathcal{D}^j$; split into minibatches of size B;
        for *each minibatch* $\mathcal{M} \subset \mathcal{D}^j$ **do**
            for *each agent* $i \in V$ **do**
                Compute importance ratios $\rho_t^i\left(\theta_{i,\lambda^j}\right)$ via (38);
                Update θ_{i,λ^j} using clipped surrogate (37) with entropy coeff c_{ent};
            end
            Compute value loss L_V via (39) with $V_{\phi_{\lambda^j}}(s_t)$; update ϕ_{λ^j} with coeff c_V;
        end
        if *average KL*$\left(\left\{\pi_{\theta_{i,old,\lambda^j}}\right\} \parallel \left\{\pi_{\theta_{i,\lambda^j}}\right\}\right)$ *exceeds target* **then**
            break
        end
    end
end
$\left\{\theta^*_{i,\lambda^j}\right\}_{i\in V} \leftarrow \{\theta_{i,\lambda^j}\}_{i\in V}$;
$\phi^*_{\lambda^j} \leftarrow \phi_{\lambda^j}$;
return $\left\{\theta^*_{i,\lambda^j}\right\}_{i\in V}$, $\phi^*_{\lambda^j}$;

samples an action from its policy based on its local observation; the joint action is, then, executed in the environment; the subsequent global and local states are observed; and the multiobjective reward vector is scalarized via the weighted-sum function $g^{\text{ws}}(\cdot \mid \lambda^j)$. The corresponding transitions are stored in $\mathcal{D}^j$. After trajectory collection, the current policies are stored as old policies for PPO-style importance sampling; the centralized critic computes bootstrapped target values and TD residuals using the global state; and per-agent advantages are estimated via GAE, optionally normalized for each agent.

Policy optimization is, then, performed over K_{ppo} epochs in each round. In every epoch, the buffer $\mathcal{D}^j$ is shuffled and partitioned into minibatches. For each minibatch, every agent updates its policy parameters using the clipped surrogate PPO objective with entropy regularization, based on its advantage estimates and importance sampling ratios. The centralized critic is updated by minimizing the value loss on the same minibatches, with global states as inputs. If the average Kullback–Leibler (KL) divergence between the old and updated policies exceeds a predefined threshold, the update loop is terminated early; otherwise, the procedure continues until the epoch finishes and proceeds to the next round, up to the limit $R_{\max}$.

Upon termination, the optimized set of independent actors $\{\pi^*_{\theta_{i,\lambda^j}}\}_{i\in V}$ and the centralized critic $V^*_{\phi_{\lambda^j}}$ are returned for subproblem j. During execution, only the decentralized actors are employed, enabling each agent to make decisions solely from its local observations, while the centralized critic is utilized exclusively during offline training. This design is well-suited for large-scale LEO satellite networks, as it reduces online communication overhead, enhances scalability, and improves robustness, while ensuring that each learned routing policy achieves the tradeoff specified by λ^j.

F. Complexity Analysis

We analyze the computational complexity of the proposed framework. For a single subproblem in Algorithm 2, during each training round, trajectory collection requires $\mathcal{O}(\mathcal{N}H|V|)$ operations, as $|V|$ agents (satellites) interact with the environment for H steps over $\mathcal{N}$ episodes. The subsequent policy and critic updates are performed across K_{ppo} epochs, each involving all $|V|$ actors and the critic, resulting in a per-round cost of $\mathcal{O}(K_{\text{ppo}}\mathcal{N}H|V|)$. Hence, the overall per-round time complexity is $\mathcal{O}(\mathcal{N}H|V|(K_{\text{ppo}}+1))$, and over at most $R_{\max}$ training rounds, the per-subproblem time complexity becomes

$$\mathcal{O}\big(R_{\max}\,\mathcal{N}H\,|V|\,\big(K_{\text{ppo}}+1\big)\big). \tag{44}$$

The corresponding space complexity per round is dominated by the trajectory buffer and model parameters, i.e.,

$$\mathcal{O}\big(\mathcal{N}H\,|V| + |V|\,P_\theta + P_\phi\big) \tag{45}$$

which simplifies to $\mathcal{O}(\mathcal{N}H|V|)$ when parameter sizes are negligible since the buffer is reinitialized in each round.

Because Algorithm 1 sequentially solves J subproblems by invoking Algorithm 2, the total time complexity is

$$\mathcal{O}\big(J\,R_{\max}\,\mathcal{N}H\,|V|\,\big(K_{\text{ppo}}+1\big)\big) \tag{46}$$

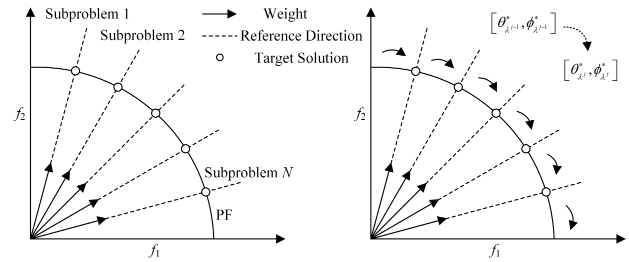


Fig. 1. Decomposition-based Pareto optimization method combined with neighborhood-based transfer learning [48].

while the space complexity remains of the same order as that of a single round for one subproblem. Overall, the framework exhibits linear scalability with respect to the number of subproblems J, the number of agents $|V|$, and the horizon length H, which makes it well-suited for parallelization in large-scale satellite network scenarios.

VI. Performance Evaluation

A. Simulation Setting

To evaluate the proposed solution, simulations were conducted using XuanCe [51], an open-source MARL framework, within a custom packet routing environment. Experiments were performed on a Mac mini equipped with 32 GB of RAM and an Apple M2 Pro 10-core CPU, using PyCharm 2023.3 Professional. Four satellite constellations from the Starlink1 network, namely, Group5, Group3, Group2, and Group4, were selected, comprising 172, 348, 720, and 1584 nodes, respectively, as detailed in Table II. Network topologies were generated using StarPerf [52], a simulator developed by SpaceNetLab at Tsinghua University. For routing evaluation, source–destination node pairs with the maximum path length (in hops) were selected under each topology to represent challenging routing scenarios.

The proposed method was benchmarked against two categories of baselines, specifically MARL-based distributed packet routing strategies and multi-objective evolutionary algorithms, which are listed as follows.

1) *GraphPR [53]:* A graph neural network-enhanced MARL-based routing algorithm for LEO networks. It integrates GAT-encoded, one-hop neighbor information to enable multihop-aware, distributed decision-making.
2) *MAFDR [54]:* A fully distributed MARL routing method in which each satellite relies solely on local spatial and queue information for decision-making.
3) *ϵ-DMOGA [55]:* A discrete multiobjective genetic algorithm that optimizes latency, spectral efficiency, and path stability.
4) *NC-MOPSO [56]:* A centrality-guided multiobjective particle swarm optimization approach designed to enhance routing efficiency in dynamic environments.

GraphPR and MAFDR are employed as baseline methods for evaluating the performance of MARL-based distributed packet routing, whereas ϵ-DMOGA and NC-MOPSO are used as benchmarks for multiobjective Pareto optimization. This two-pronged evaluation strategy enables a comprehensive

TABLE I
COMPARISON BETWEEN THIS WORK AND EXISTING RESEARCH, WHERE THE SYMBOL (✓) INDICATES THAT THE TOPIC IS A PRIMARY FOCUS

Ref.	Satellite Networks	Agent Type	Underlying Algorithm	Queueing Model	Multi-objective optimization	Pareto Optimization
[21]		Single	Q-learning			
[22]		Single	DQN			
[23]		Multiple	Independent-DQN			
[12]		Multiple	meta-MAPPO		✓	
[24]	✓	Multiple	Independent-DQN		✓	
[13]	✓	Multiple	Independent-DQN	M/M/1/m	✓	
[25]		Multiple	Not Specified	Queueing Network	✓	
[26]	✓	Multiple	Independent-DDQN	M/D/C/N	✓	
[27]		Multiple	Conjectural-MAQL	Geo/Geo/1	✓	
[28]		Single	DDPG	M/M/C/N		✓
[29]–[31]						✓
[15]	✓	Multiple	Constrained MARL		✓	
Our Work	✓	Multiple	MAPPO with GATs	G/G/1/K	✓	✓

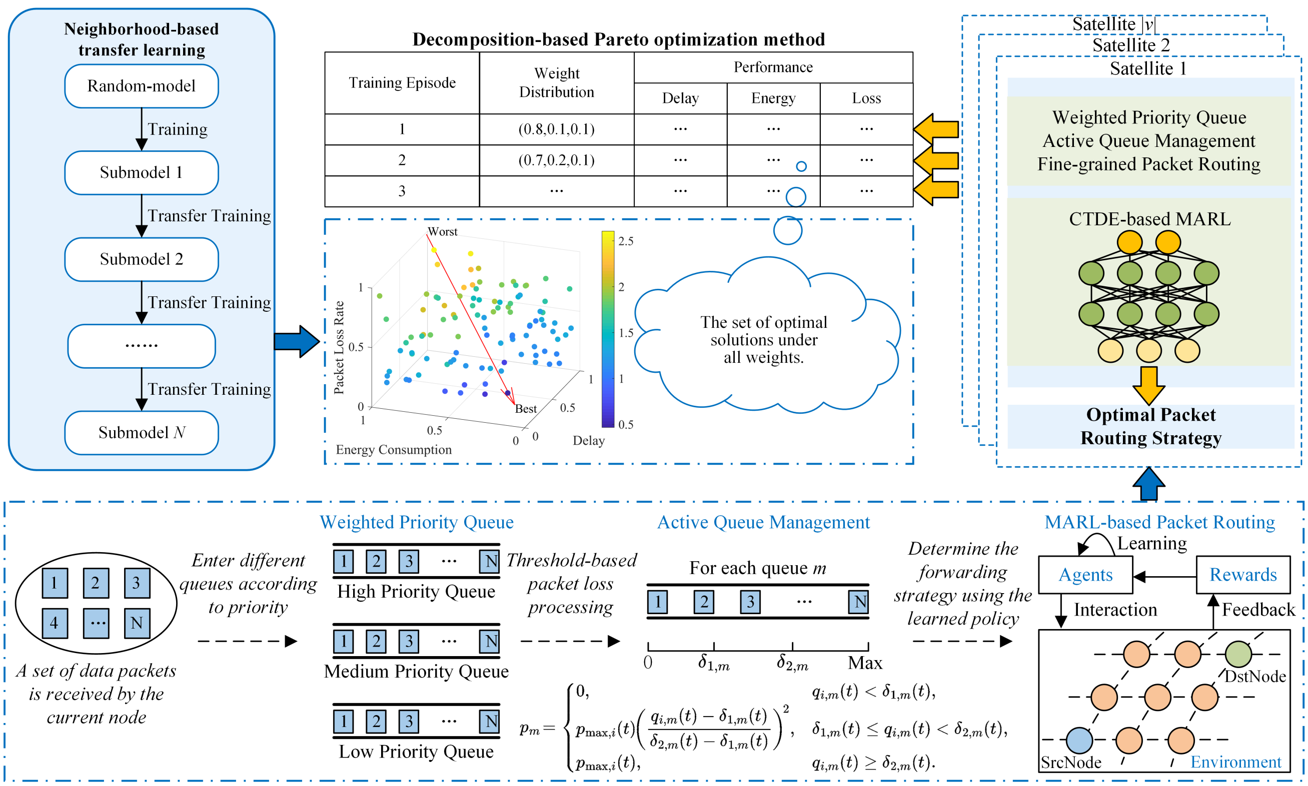


Fig. 2. Framework of the proposed Pareto-optimal multiobjective MARL packet routing optimization method.

assessment of the proposed MARL-based and Pareto-driven framework. Classical centralized shortest-path and heuristic protocols are not treated as primary baselines in this study because they typically rely on global state and single-criterion formulations, whereas our setting is decentralized, partially observable, and explicitly multiobjective. We, therefore, focus on MARL-based distributed routing and multiobjective evolutionary optimizers that better align with our problem formulation. The key simulation parameters are summarized in Table III, and detailed numerical results are provided in Section VI-B.

B. Numerical Results

We first examine the convergence behavior of the proposed POMAP method by comparing its episode rewards with those of two DRL/MARL-based routing baselines, GraphPR and MAFDR. As shown in Fig. 4(a)–(d), GraphPR achieves faster initial convergence, reaching near-maximum reward levels within 2×10^5 steps. However, its performance exhibits frequent oscillations thereafter, with amplitude fluctuations exceeding 1.5×10^4, particularly in the 348- and 720-node cases. This instability suggests potential issues such as policy overfitting or sensitivity to environmental transitions. In contrast, MAFDR yields significantly lower reward values across all node scales, with average episode rewards rarely exceeding 5×10^4, and a substantial spread throughout training. This indicates limited learning capability and poor adaptability to dynamic satellite topologies. POMAP converges more gradually, typically requiring around 3.5×10^5 steps, but ultimately achieves a peak episode reward of approximately 1.55×10^5, marginally surpassing GraphPR. More importantly, POMAP exhibits a stable learning trajectory with low variability,

TABLE II
SATELLITE CONSTELLATION PARAMETERS IN SIMULATION

Satellite Constellation	Satellite Altitude	Number of Satellites	Number of Planes	Satellites Per Plane
Starlink1 Group5	560 km	172	4	43
Starlink1 Group3	570 km	348	6	58
Starlink1 Group2	540 km	720	36	20
Starlink1 Group4	550 km	1584	72	22

TABLE III
SIMULATION PARAMETERS AND NOTATIONS

Symbol	Description	Value/Range
Δt	Slot duration	1 s
B	Channel bandwidth	10 GHz
f	Carrier frequency	30 GHz
c	Speed of light	3×10^8 m/s
S	Packet size	1500 Byte
N	Total packets	10000 packets
K	Total queue capacity per node	300 packets
M	Number of traffic queues	3
K_m	Queue capacity per class	$K/3$
$p_{\max}$	Max drop probability	0.1
P_{tx}	Transmit power	5 W
P_{rx}	Receive power	1 W
E_{init}	Initial energy	100 J
D_{max}	Maximum allowed end-to-end delay	200 ms

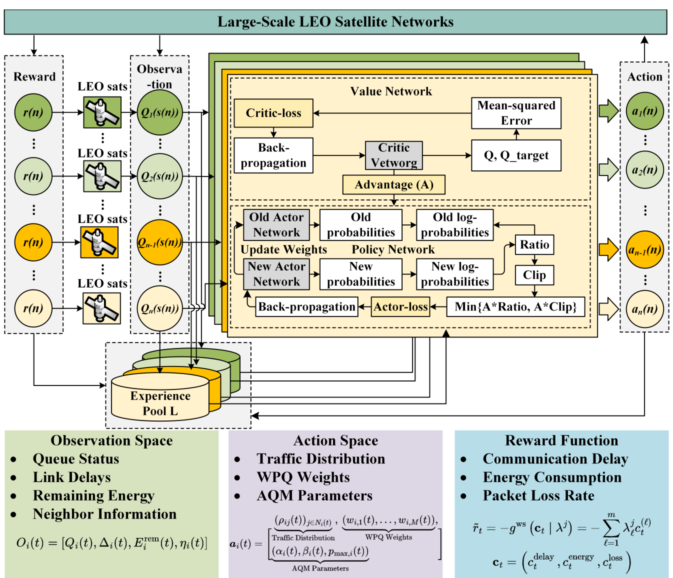


Fig. 3. Proposed MAPPO-based packet routing method for large-scale satellite networks.

especially in large-scale scenarios [Fig. 4(c) and (d)]. This stability reflects the robustness of its policy updates and the effectiveness of the MAPPO-based optimization framework in handling partially observable and nonstationary network environments. All reported curves represent the mean over five independent runs to ensure statistical reliability.

Building upon this convergence analysis, we next evaluate the multiobjective optimization capability of POMAP across different network scales. Fig. 5(a)–(d) illustrates the Pareto fronts in terms of average packet delay (ms), total energy consumption (J), and packet loss rate, where lower values on all axes are desirable. Across all network sizes, POMAP consistently generates Pareto fronts that span a broader region in the objective space and exhibit smoother distribution patterns compared to ϵ-DMOGA and NC-MOPSO. In smaller networks (172 and 348 nodes), POMAP's solutions cover a more extensive area across all objectives, achieving lower delay and packet loss while maintaining comparable energy usage. The fronts of ϵ-DMOGA and NC-MOPSO appear more clustered and less uniformly distributed, particularly along the loss-rate axis. As the network scales increase to 720 and 1584 nodes, the contrast becomes more pronounced. POMAP maintains a well-spread and diverse Pareto surface, whereas ϵ-DMOGA clusters toward suboptimal regions with high energy and loss values. NC-MOPSO achieves moderate dispersion but still lags behind POMAP in front coverage and uniformity. These results confirm POMAP's scalability and its ability to generate balanced tradeoff solutions under increasing network complexity.

To further validate the effectiveness of the proposed method, we evaluate the quality of the obtained Pareto solutions using four widely used metrics in multi-objective optimization: the number of non-dominated solutions (NDS), hypervolume, diversity, and spread. These metrics jointly assess convergence accuracy, solution coverage, and distribution balance, and are presented in Fig. 6(a)–(d). As shown in Fig. 6(a), POMAP consistently achieves higher NDS values across all network sizes. Compared to the best-performing baseline, POMAP offers over 150% improvement in smaller networks and maintains more than 100% advantage as the node count increases, demonstrating superior capacity to discover a broad set of tradeoff solutions. Correspondingly, the hypervolume results in Fig. 6(b) reveal that POMAP dominates the objective space more extensively. Its hypervolume exceeds that of the baselines by more than 100% in all cases, and the performance gap grows with increasing network size, highlighting POMAP's scalability and convergence strength. In terms of diversity [Fig. 6(c)], POMAP maintains high and consistent values across all scenarios. While NC-MOPSO shows slightly higher diversity at the largest scale, POMAP's diversity remains more stable. In contrast, ϵ-DMOGA consistently underperforms with significantly lower diversity values, indicating poor solution spread and exploration. The spread results [Fig. 6(d)] further confirm POMAP's superiority. Its spread is 60%–70% smaller than ϵ-DMOGA and marginally lower than NC-MOPSO in most settings, suggesting that POMAP generates more uniformly distributed Pareto solutions with minimal clustering.

Following the multiobjective assessment, we now examine single-objective metrics: communication delay, energy consumption, and packet loss rate. Figs. 7–9 report these metrics across two large-scale networks with 720 and 1584 nodes. For communication delay (Fig. 7), POMAP consistently achieves lower median latency compared to both baselines. At 720 nodes, the median delay is reduced by approximately 9% relative to ϵ-DMOGA and by about 6% compared to NC-MOPSO. Similar improvements are observed at 1584 nodes. Furthermore, POMAP maintains a narrower spread in delay distribution, indicating more stable and predictable performance under dynamic conditions. In terms of energy consumption (Fig. 8), POMAP demonstrates notable

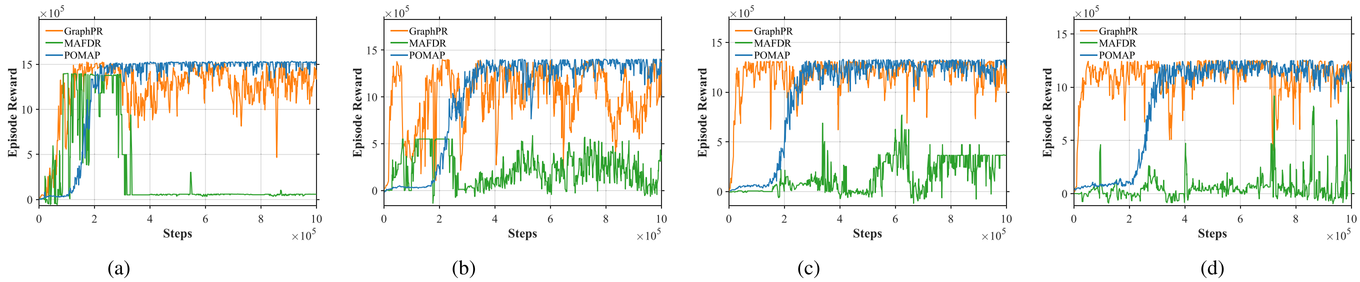


Fig. 4. Average episode rewards of POMAP, GraphPR, and MAFDR under different Starlink1 network sizes. Results are averaged over five independent runs. (a) 172 nodes. (b) 348 nodes. (c) 720 nodes. (d) 1584 nodes.

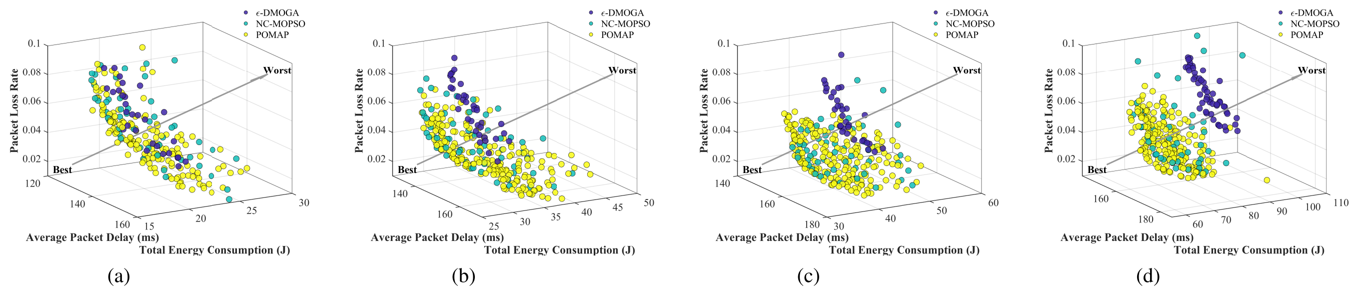


Fig. 5. 3-D Pareto fronts obtained by POMAP, ϵ-DMOGA, and NC-MOPSO on Starlink1 networks of varying sizes. Objectives include average delay (ms), energy consumption (J), and packet loss rate. Lower values on all axes indicate better performance. (a) 172 nodes. (b) 348 nodes. (c) 720 nodes. (d) 1584 nodes.

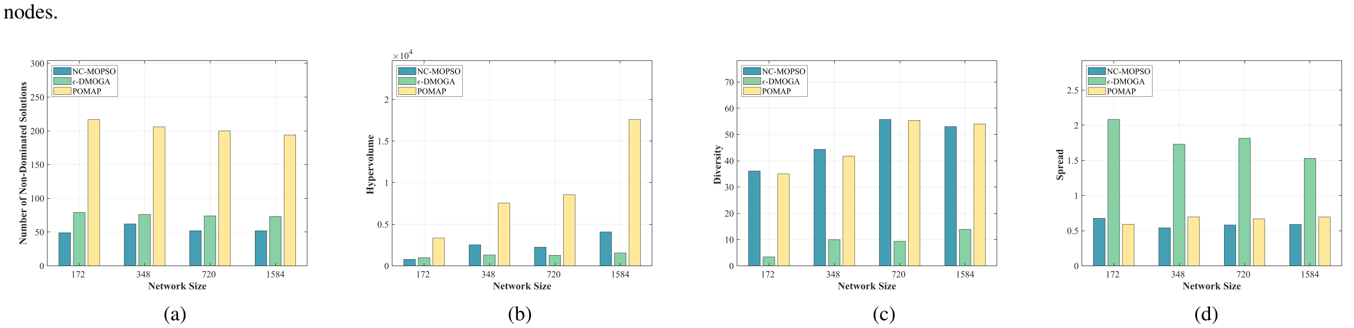


Fig. 6. Comparison of multiobjective performance metrics for POMAP, ϵ-DMOGA, and NC-MOPSO across different network sizes: (a) number of NDSs; (b) hypervolume; (c) diversity; and (d) spread. Higher NDS, hypervolume, diversity, and lower spread are desirable.

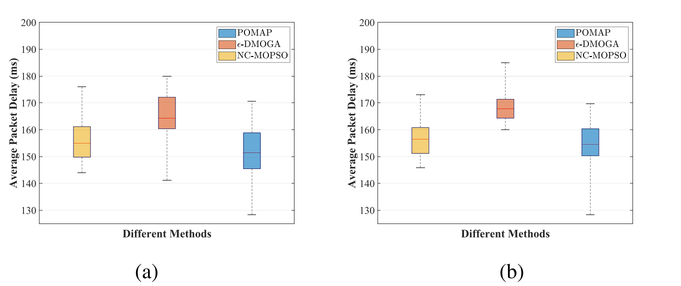


Fig. 7. Comparison of average packet delay performance metrics across large-scale satellite networks: (a) 720 nodes and (b) 1584 nodes.

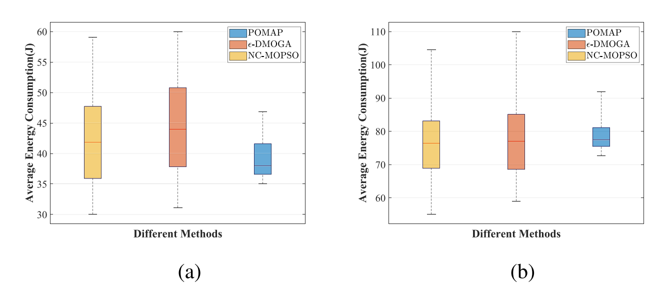
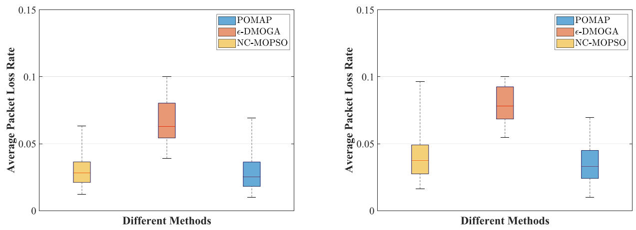


Fig. 8. Comparison of average energy consumption performance metrics across large-scale satellite networks: (a) 720 nodes and (b) 1584 nodes.

efficiency. At 720 nodes, it reduces median energy usage by approximately 15% compared to ϵ-DMOGA and by 8% compared to NC-MOPSO. The improvement becomes more substantial at 1584 nodes, with reductions up to 17%. In both cases, the spread of energy usage remains tighter under POMAP, suggesting better control over power efficiency in large-scale environments. Regarding packet loss rate (Fig. 9), POMAP again outperforms the baselines. At 720 nodes, it reduces median loss by over 30% compared to ϵ-DMOGA and around 15% relative to NC-MOPSO. At 1584 nodes, the advantage becomes even more evident. The compact spread and lack of outlier deviations further confirm the robustness of POMAP in ensuring reliable data transmission.



(a) (b)

Fig. 9. Comparison of average packet loss rate performance metrics across large-scale satellite networks: (a) 720 nodes and (b) 1584 nodes.

In summary, POMAP exhibits robust and scalable performance across multiple evaluation metrics and network sizes, achieving stable convergence and well-balanced tradeoffs. These characteristics underscore its practical potential for deployment in heterogeneous satellite constellations and for integration with the quality of experience-aware or priority-based routing strategies in future large-scale network systems. Nevertheless, practical deployment in large-scale LEO constellations raises additional challenges, including onboard computational constraints, hardware limitations, and the need for real-time, fault-tolerant operation under rapidly changing topologies. Addressing these challenges through model compression, parameter sharing, lightweight neural architectures, and hardware acceleration, as well as partial ground-based processing, remains an important direction for future work.

VII. Conclusion

This article addresses the challenges of distributed packet routing in large-scale LEO satellite networks, where dynamic topologies, constrained bandwidth, and partial observability significantly hinder the effectiveness of conventional methods. To this end, we propose POMAP, a Pareto-optimal MARL framework tailored for multiobjective packet routing. The framework incorporates a G/G/1/K queueing model and a dynamic multiattribute graph representation to capture node-level queueing behavior, energy consumption, and time-varying link conditions. The inherently conflicting objectives of delay minimization, energy efficiency, and packet loss reduction are jointly optimized by decomposing the problem into scalar subproblems, each addressed with a MAPPO-based policy learning mechanism. Simulation results on realistic satellite network topologies confirm that the proposed approach effectively alleviates network congestion, improves transmission performance, and enhances routing robustness under highly dynamic conditions. These results demonstrate the potential of integrating Pareto-based MARL with queue-aware modeling for scalable, adaptive, and QoS-sensitive packet routing in the next-generation satellite communication systems.

Future work may explore enhancing the adaptability and scalability of the POMAP framework. One promising direction is to integrate meta-learning or transfer learning techniques to enable faster policy adaptation across varying traffic patterns and satellite topologies, particularly when extending to larger constellations or heterogeneous mission profiles. Another avenue involves advancing the queue-aware decision-making mechanism, such as integrating delay-sensitive or QoS-aware traffic classification into the G/G/1/K model, thereby improving routing precision under mixed service demands. Additionally, deploying lightweight onboard inference modules using AI accelerators may enable real-time decision-making with minimal reliance on ground control, further supporting autonomous packet routing in resource-constrained satellite environments. These directions aim to further extend the applicability of POMAP to real-world satellite systems operating under diverse and evolving conditions.

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